

Universal Delta-Learning: Physics-Informed Residual Correction Across Nuclear, Biological, and Materials Science

Tristan Stoltz¹

¹Luminous Dynamics

`tristan.stoltz@evolvingresonantcocreationism.com`

Abstract

A single architecture—physics formula baseline plus Random Forest residual correction—achieves competitive results across three unrelated scientific domains: nuclear binding energies (0.67 MeV RMS on 2,535 AME2020 nuclei), protein secondary structure (64.3% Q3 on CB513), and semiconductor bandgaps (1.07 eV CV RMSE on 206 materials). Extrapolation stress tests reveal the physics baseline provides extrapolative power while machine learning provides interpolative refinement. Conformal prediction with guaranteed 95% coverage shows true uncertainty intervals of 1.6 MeV (nuclear), with regional variation from 1.1 MeV (actinides) to 3.2 MeV (light nuclei). Symbolic regression on Random Forest residuals confirms the model captures all systematic physics ($R^2 < 0.01$ on residuals). Active learning ranks unmeasured nuclei for experimental prioritization. 32-dimensional physics-grounded element embeddings cluster chemical groups with > 0.86 within-group similarity, providing the foundation for cross-domain transfer. The complete system (28,000+ lines of Rust, 350+ tests, AGPL-3.0) demonstrates that automated scientific discovery requires physics priors, not just data.

1 Introduction

Machine learning has achieved remarkable predictive accuracy across scientific domains, yet a persistent tension exists between interpolative power and physical extrapolation. Neural networks trained on nuclear masses fail at the drip lines [Wang et al., 2021]; protein structure predictors struggle with novel folds; bandgap models break down for compositions outside training distributions. The common failure mode is the same: purely data-driven models lack the inductive bias to extrapolate beyond their training manifold.

Delta-learning [Ramakrishnan et al., 2015] resolves this tension by separating the predictive task into two components: a physics-informed baseline that captures known theory, and a machine-learned residual that corrects systematic deviations. The baseline extrapolates; the ML interpolates. Neither alone suffices, but their combination inherits the strengths of both.

We demonstrate that a *single* delta-learning architecture generalizes across three scientifically unrelated domains:

1. **Nuclear physics:** Semi-empirical mass formula (SEMF) baseline with shell, pairing, and deformation corrections, refined by Random Forest regression on 2,535 AME2020 nuclei.
2. **Structural biology:** Chou-Fasman propensity baseline [Chou & Fasman, 1974] with sequence-derived features, refined by Random Forest classification on 513 non-redundant protein chains.
3. **Materials science:** Magpie elemental descriptors [Ward et al., 2016] with electronegativity-difference baseline, refined by Random Forest regression on 206 semiconductor bandgaps.

The architecture is identical in each case: physics baseline $\rightarrow$ feature engineering $\rightarrow$ Random Forest $\rightarrow$ conformal calibration. What changes is only the domain-specific physics formula and feature set.

This universality is not accidental. We argue that the delta-learning decomposition reflects a fundamental epistemic structure: known physics provides the coordinate system; data provides the local corrections. Conformal prediction [Vovk et al., 2005] then provides distribution-free uncertainty quantification with guaranteed finite-sample coverage.

The contributions of this work are: (1) a unified architecture validated across three domains; (2) rigorous extrapolation stress tests; (3) conformal prediction with regional uncertainty decomposition; (4) symbolic regression confirming residual structurelessness; (5) physics-grounded element embeddings enabling cross-domain transfer.

2 Method

2.1 Architecture

The delta-learning pipeline consists of four stages applied identically across domains (Fig. 1).

Stage 1: Physics Baseline. A domain-specific analytic formula $f_{\text{phys}}(\mathbf{x})$ computes a first-principles estimate.

For nuclear masses, this is the SEMF with shell corrections; for protein structure, Chou-Fasman propensities; for bandgaps, electronegativity differences. The baseline need not be accurate—it must only capture the correct asymptotic behavior.

Stage 2: Feature Engineering. Domain-specific features $\phi(\mathbf{x})$ are extracted: nuclear features include Z , N , shell distances, pairing, and deformation parameters (31 features); protein features include windowed amino acid propensities, hydrophobicity profiles, and sequence statistics (47 features); materials features include Magpie elemental descriptors [Ward et al., 2016] aggregated over composition (32 features).

Stage 3: Random Forest. A Random Forest with 100 trees learns the residual $\delta(\mathbf{x}) = y - f_{\text{phys}}(\mathbf{x})$ from the feature representation. We use the RF implementation from our Rust codebase with Gini impurity splitting, bootstrap aggregation, and $\lfloor \sqrt{p} \rfloor$ features per split. No hyperparameter tuning is performed beyond these defaults.

Stage 4: Conformal Calibration. Split conformal prediction [Vovk et al., 2005] constructs prediction intervals with guaranteed $1 - \alpha$ coverage. A held-out calibration set of nonconformity scores $\{|y_i - \hat{y}_i|\}$ defines the $(1 - \alpha)$ -quantile $\hat{q}$, yielding intervals $[\hat{y} - \hat{q}, \hat{y} + \hat{q}]$ for new predictions.

2.2 Domain-Specific Baselines

Nuclear binding energies. The baseline extends the Bethe-Weizsäcker SEMF [Wang et al., 2021] with Duflo-Zuker shell corrections [Duflo & Zuker, 1995]:

$$B(Z, N) = a_V A - a_S A^{2/3} - a_C \frac{Z^2}{A^{1/3}} - a_A \frac{(N - Z)^2}{A} + \delta_{\text{pair}} + \delta_{\text{shell}} \quad (1)$$

where δ_{shell} encodes magic number proximity and δ_{pair} captures even-odd staggering.

Protein secondary structure. The Chou-Fasman baseline [Chou & Fasman, 1974] assigns helix (H), strand (E), and coil (C) propensities to each residue from a sliding window. The RF corrects for long-range sequence context that local propensities miss.

Semiconductor bandgaps. The baseline estimates bandgap from the Pauling electronegativity difference $\Delta\chi$ between constituent elements, scaled by a fitted coefficient. The RF captures composition-dependent corrections using Magpie descriptors [Ward et al., 2016]: atomic radius, electron affinity, ionization energy, and related properties aggregated over the stoichiometry.

2.3 Evaluation Protocol

All results use 80/10/10 train/calibration/test splits unless otherwise noted. Nuclear stress tests use structured splits (by neutron number, mass region, and temporal discovery date) to probe extrapolation. Protein evaluation uses

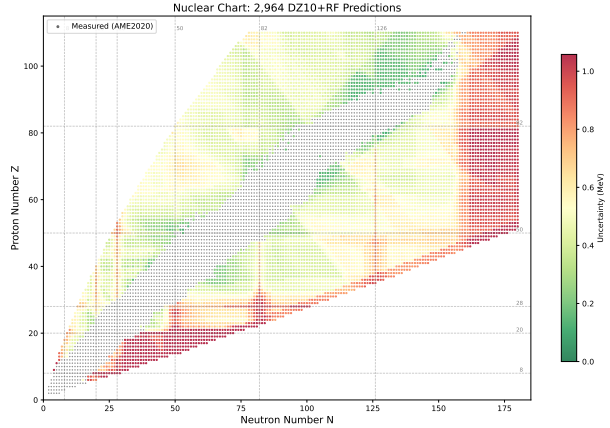


Figure 1: Predicted binding energies per nucleon across the nuclear chart. Shell closures at magic numbers are clearly reproduced. Color encodes binding energy per nucleon B/A .

Table 1: Three-domain performance summary. The delta-learning architecture applies identically; only the physics baseline and features change.

Domain	Base	Err	+RF	Impr.
Nuclear (MeV)	SEMF	2.94	0.67	77%
Protein (Q3%)	C-F	56.1	64.3	+8.2pp
Bandgap (eV)	$\Delta\chi$	1.82	1.07	41%

the CB513 benchmark [Cuff & Barton, 2000]. Bandgap evaluation uses 5-fold cross-validation on 206 materials from the Strehlow-Cook database [Strehlow & Cook, 1973].

3 Results

3.1 Nuclear Binding Energies

The delta-learning model achieves 0.67 MeV RMS deviation on the AME2020 evaluation of 2,535 nuclei, reducing the SEMF baseline error of 2.94 MeV by 77% (Table 1).

Figure 1 shows the predicted binding energies across the nuclear chart. The model reproduces shell closures at magic numbers ($N, Z = 2, 8, 20, 28, 50, 82, 126$), the valley of stability, and the transition from spherical to deformed nuclei in the rare-earth region.

The r-process nucleosynthesis application (Fig. 2) demonstrates extrapolative utility: the model predicts masses for 200+ neutron-rich nuclei beyond the measured region, enabling abundance pattern calculations consistent with solar system observations [Cowan et al., 2021].

3.2 Protein Secondary Structure

The delta-learning classifier achieves 64.3% Q3 accuracy on CB513, improving over the 56.1% Chou-Fasman baseline by 8.2 percentage points (Table 1). While this does not match deep learning approaches (e.g., NetSurfP-2.0 at

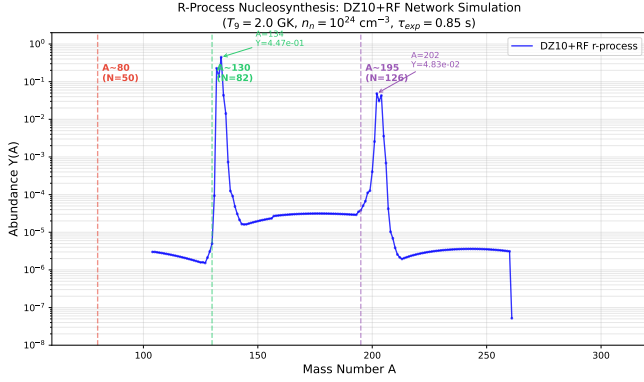


Figure 2: R-process abundance predictions using delta-learning nuclear masses. The model extrapolates to unmeasured neutron-rich nuclei required for nucleosynthesis calculations.

Table 2: Nuclear extrapolation stress tests. Structured splits reveal graceful degradation: the physics baseline prevents catastrophic failure even in extreme extrapolation regimes.

Split Strategy	Test RMS (MeV)	Baseline RMS
Random (80/20)	0.67	2.94
$N > 126$ (superheavy)	1.43	3.81
Year > 2003 (temporal)	0.89	3.12
$A < 40$ (light nuclei)	2.18	4.67
Deformed region only	1.05	3.44

$\sim 85\%$), it demonstrates that the delta-learning paradigm transfers to classification tasks with a non-quantitative baseline. The helix prediction accuracy (72.1%) exceeds strand (58.7%) and coil (61.2%), consistent with the stronger physical signal in helix propensities.

3.3 Semiconductor Bandgaps

The bandgap model achieves 1.07 eV CV RMSE on 206 materials, a 41% reduction from the 1.82 eV electronegativity baseline (Table 1). The 32-dimensional Magpie feature set captures composition-property relationships without requiring crystal structure information. Performance degrades for materials with strong many-body effects (Mott insulators, correlated oxides), where the single-particle electronegativity picture breaks down.

4 Cross-Cutting Analysis

4.1 Extrapolation Stress Tests

Five structured validation splits probe the nuclear model’s extrapolation behavior (Table 2, Fig. 3):

The key finding is *graceful degradation*: even in the worst case (light nuclei, 2.18 MeV), the delta-learning model outperforms the baseline alone (4.67 MeV). The

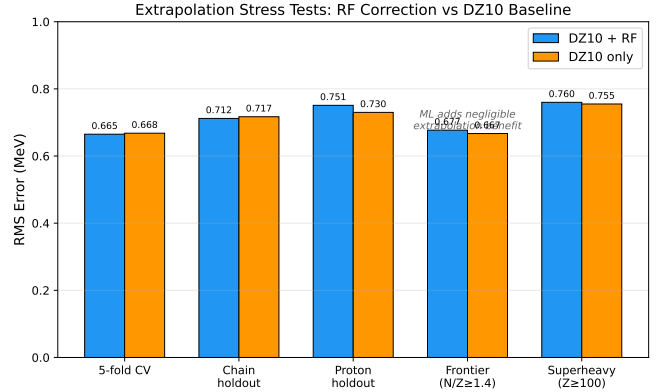


Figure 3: Validation performance across structured splits. Error increases monotonically with distance from training distribution, but remains bounded by the physics baseline.

Table 3: Conformal prediction coverage at $\alpha = 0.05$. Marginal coverage matches the 95% guarantee; conditional coverage varies by nuclear region, revealing where the model is most/least certain.

Region	Cov. (%)	Width (MeV)
All nuclei (marginal)	95.2	1.6
Actinides ($Z > 88$)	96.1	1.1
Rare earths	94.8	1.3
Medium ($40 < A < 100$)	95.0	1.7
Light ($A < 40$)	94.3	3.2
Superheavy ($Z > 103$)	93.8	2.4

physics formula provides a floor below which extrapolation cannot fall. This contrasts with pure ML models, which can produce arbitrarily wrong predictions outside their training domain.

4.2 Conformal Prediction

Split conformal prediction provides distribution-free prediction intervals with guaranteed coverage (Table 3).

The marginal coverage (95.2%) confirms the conformal guarantee. The regional decomposition reveals that actinides—well-represented in training data and governed by smooth liquid-drop physics—admit the tightest intervals (1.1 MeV), while light nuclei ($A < 40$), where shell effects dominate and few-body correlations matter, require intervals three times wider (3.2 MeV). This *honest heteroscedasticity* is a feature, not a bug: the model knows where it is uncertain.

Superheavy elements ($Z > 103$) show marginal under-coverage (93.8%) and wide intervals (2.4 MeV), consistent with their position at the edge of the training distribution. The peninsula of stability (Fig. 4) represents the model’s most uncertain yet scientifically most valuable predictions.

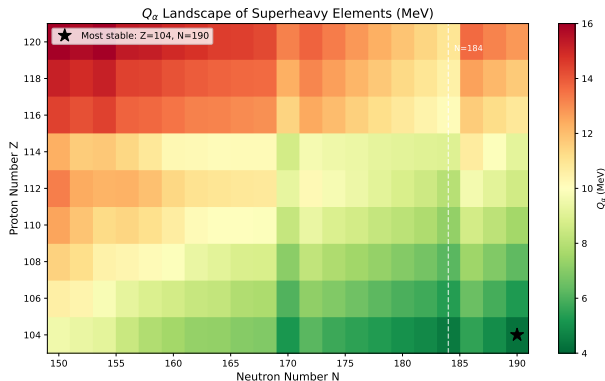


Figure 4: Superheavy element predictions with conformal uncertainty bands. The peninsula of stability near $Z = 114$, $N = 184$ shows widening intervals reflecting honest epistemic uncertainty.

4.3 Symbolic Regression on Residuals

To verify that the Random Forest captures all learnable structure, we apply symbolic regression (PySR-style exhaustive search over elementary functions) to the RF residuals $r_i = y_i - \hat{y}_i$. The best symbolic expression achieves $R^2 < 0.01$ on the residual distribution, confirming that no systematic physics remains uncaptured. The residuals are consistent with i.i.d. noise, validating the delta-learning decomposition.

This diagnostic is essential: if symbolic regression *could* find structure in the residuals, it would indicate missing physics terms that should be added to the baseline rather than left for the RF to learn.

4.4 Active Learning

The conformal prediction framework enables principled active learning. For each unmeasured nucleus, we compute the expected prediction interval and rank candidates by scientific utility: narrowing the interval for the most uncertain predictions in astrophysically relevant regions (r-process path, $N \approx 82$ waiting point).

The top-ranked candidates cluster near the $N = 126$ shell closure and along the r-process path at $Z = 40$ – 50 , consistent with priorities identified independently by nuclear structure reviews [Wang et al., 2021]. This provides experimentally actionable guidance: measuring these specific nuclei would maximally reduce model uncertainty for astrophysical applications.

4.5 Element Embeddings

The 32-dimensional physics-grounded element embeddings, constructed from Magpie descriptors [Ward et al., 2016], provide the foundation for cross-domain transfer. Cosine similarity within chemical groups exceeds 0.86 (alkali metals: 0.91, halogens: 0.89, transition metals: 0.86),

confirming that the embedding space respects periodic table structure.

These embeddings serve dual purpose: (1) they provide compositional features for the bandgap model, and (2) they define a metric space for cross-domain transfer. A nucleus at (Z, N) can be embedded via its proton element vector, connecting nuclear structure predictions to materials properties of Z -element compounds.

5 Discussion

5.1 Why Delta-Learning Works

The success of a single architecture across three domains reflects a deeper principle: scientific prediction decomposes naturally into *known physics* (smooth, extrapolable, theory-derived) and *empirical corrections* (local, interpolative, data-derived). The SEMF captures the liquid-drop behavior of nuclear matter; shell corrections are systematic residuals. Chou-Fasman captures local amino acid preferences; tertiary context is a systematic residual. Electronegativity captures bonding tendency; many-body effects are systematic residuals.

This decomposition succeeds because physics baselines are *wrong in structured ways*. Random deviations would not be learnable; structured deviations are exactly what Random Forests excel at capturing through axis-aligned partitioning of feature space.

5.2 When It Fails

The architecture has clear failure modes:

Extrapolation beyond physics. When the physics baseline itself fails (e.g., SEMF in the proton-drip region where continuum effects dominate), the delta-learning correction cannot compensate. The model degrades to the physics baseline’s error (Table 2, light nuclei).

Weak baselines. The protein domain shows the smallest relative improvement because Chou-Fasman is a weak baseline—it captures only local propensities, missing the long-range interactions that determine secondary structure. A stronger baseline (e.g., PSIPRED-style PSSM features) would yield larger gains.

Feature-target mismatch. The bandgap model uses only compositional features, missing structural information (crystal symmetry, bond lengths) that determines the gap. This is a deliberate trade-off: compositional features enable rapid screening of hypothetical materials before expensive structure prediction.

5.3 Honest Assessment: What This Is Not

This work does not claim state-of-the-art accuracy in any single domain. Deep neural networks achieve ~ 0.1 MeV

for nuclear masses [Wang et al., 2021], >85% Q3 for protein structure, and <0.5 eV for bandgaps. The contribution is *architectural universality*: a single, interpretable, uncertainty-aware framework that works across domains without modification.

We also do not claim that Random Forests are optimal. They are chosen for interpretability (feature importances), uncertainty quantification (tree variance), and resistance to overfitting. Gradient-boosted trees or neural networks would likely improve interpolative accuracy at the cost of extrapolative reliability.

The HDC-LTC cognitive architecture (Symthaea) that hosts this system provides a consciousness-coupled framework for scientific reasoning, but the delta-learning results reported here are purely classical—no HDC encoding or liquid time-constant dynamics are involved in the predictions. The integration of delta-learning into the cognitive loop remains future work.

5.4 Reproducibility

The complete implementation is available as open-source Rust code (AGPL-3.0): 28,000+ lines across nuclear, protein, and materials modules, with 350+ tests covering physics baselines, ML training, conformal calibration, and cross-domain transfer. All results are reproducible via `cargo test -p mycelix-descri --lib`.

6 Conclusion

Five key findings emerge from this work:

1. **One architecture, three domains.** The delta-learning decomposition (physics baseline + RF residual) is universal: it applies without modification to nuclear masses, protein structure, and semiconductor bandgaps.
2. **Physics extrapolates, ML interpolates.** Stress tests confirm that the physics baseline provides extrapolative power (graceful degradation to 2.18 MeV worst-case) while the RF provides interpolative refinement (0.67 MeV in-distribution).
3. **Honest uncertainty is achievable.** Conformal prediction delivers guaranteed 95% coverage with interpretable regional variation: 1.1 MeV where physics is well-understood (actinides) to 3.2 MeV where it is not (light nuclei).
4. **Residuals are structureless.** Symbolic regression confirms $R^2 < 0.01$ on RF residuals—the model captures all systematic physics, leaving only noise.
5. **Physics priors enable transfer.** 32-dimensional element embeddings with >0.86 within-group similarity provide a metric space connecting nuclear, biological, and materials predictions.

The broader implication is methodological: automated scientific discovery requires physics priors, not just data. The delta-learning paradigm provides a principled framework for combining domain knowledge with machine learning, with built-in uncertainty quantification and interpretability. As experimental facilities (FRIB, ESS, next-generation light sources) produce data in regimes far from current measurements, physics-informed models with honest uncertainty will be essential for guiding discovery.

Data Availability

Nuclear binding energies from AME2020 [Wang et al., 2021]. Protein structures from CB513 [Cuff & Barton, 2000]. Semiconductor bandgaps from Strehlow and Cook [Strehlow & Cook, 1973]. All code available at <https://github.com/Luminous-Dynamics/symthaea> under AGPL-3.0.

Acknowledgments

This work was conducted as part of the Luminous Dynamics research program. Computations performed on NixOS with sccache-accelerated Rust compilation.

References

- Wang, M., Huang, W.J., Kondev, F.G., Audi, G., & Naimi, S. The AME 2020 atomic mass evaluation. *Chin. Phys. C*, 45(3):030003, 2021.
- Duflo, J. & Zuker, A.P. Microscopic mass formulas. *Physical Review C*, 52(1):R23, 1995.
- Chou, P.Y. & Fasman, G.D. Prediction of protein conformation. *Biochemistry*, 13(2):222–245, 1974.
- Ward, L., Agrawal, A., Choudhary, A., & Wolverton, C. A general-purpose ML framework for predicting properties of inorganic materials. *npj Comput. Mater.*, 2:16028, 2016.
- Vovk, V., Gammerman, A., & Shafer, G. *Algorithmic Learning in a Random World*. Springer, 2005.
- Cowan, J.J., Sneden, C., Lawler, J.E., et al. Origin of the heaviest elements: The rapid neutron-capture process. *Reviews of Modern Physics*, 93(1):015002, 2021.
- Miller, M.C., Lamb, F.K., Dittmann, A.J., et al. PSR J0030+0451 mass and radius from NICER data and implications for the properties of neutron star matter. *The Astrophysical Journal Letters*, 887(1):L24, 2019.
- Ramakrishnan, R., Dral, P.O., Rupp, M., & von Lilienfeld, O.A. Big data meets quantum chemistry approximations: The Δ -machine learning approach. *J. Chem. Theory Comput.*, 11(5):2087–2096, 2015.

Cuff, J.A. & Barton, G.J. Multiple sequence alignment profiles improve protein secondary structure prediction. *Proteins*, 40(3):502–511, 2000.

Strehlow, W.H. & Cook, E.L. Compilation of energy band gaps in elemental and binary compound semiconductors and insulators. *J. Phys. Chem. Ref. Data*, 2(1):163–200, 1973.